

2D Heat Diffusion Simulation

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1 Introduction

This report presents the results of a numerical simulation of two-dimensional heat diffusion in a confined space. The physical problem under consideration is the evolution of temperature in a square room containing a heat source. The room is subject to various boundary conditions that model heat exchange with the external environment through walls and a window.

The governing equation for this physical phenomenon is the heat equation, a parabolic partial differential equation. In two dimensions, it is given by:

$$\frac{\partial T}{\partial t} = \alpha \left(\frac{\partial^2 T}{\partial x^2} + \frac{\partial^2 T}{\partial y^2} \right) + S(x, y, t)$$

where $T(x, y, t)$ is the temperature at a point (x, y) and time t , α is the thermal diffusivity of the medium, and $S(x, y, t)$ is a source term, representing the heater in this simulation. The simulation aims to solve this equation numerically to observe the temperature dynamics under different physical conditions.

2 Numerical Method

To solve the heat equation numerically, a finite-difference approach was employed. The spatial domain was discretized into a uniform grid, and the time evolution was handled using the Crank-Nicolson method. This method is an implicit scheme known for its unconditional stability, which allows for larger time steps compared to explicit methods without compromising the stability of the solution.

The Crank-Nicolson scheme approximates the time derivative using a central difference at the midpoint of a time step, and averages the spatial derivatives at the beginning and end of the time step. This results in a system of linear equations for the temperatures at the next time step. The discretized equation can be written as:

$$T_{i,j}^{n+1} - T_{i,j}^n = \frac{\alpha \Delta t}{2(\Delta x)^2} \left(\delta_x^2 T_{i,j}^{n+1} + \delta_x^2 T_{i,j}^n + \delta_y^2 T_{i,j}^{n+1} + \delta_y^2 T_{i,j}^n \right)$$

where δ_x^2 and δ_y^2 are the central difference operators for the spatial second derivatives.

This system of equations was solved at each time step using the Gauss-Seidel iterative method. The boundary conditions, which model the heat transfer at the walls and window (Neumann-type boundary conditions), were incorporated into the system. The heater's operation was also controlled, in some tasks, by a thermostat logic based on the temperature at a designated sensor location.

3 Results

The simulation was executed for six different scenarios, each representing a unique physical configuration. The results, including heat maps at various time steps and temperature profiles at a sensor location, are presented below.

3.1 Scenario with Perfect Insulation

In this scenario, the room is perfectly insulated from the outside ($h = h_w = 0$), and the heater operates continuously. This represents an ideal case with no energy loss to the environment.

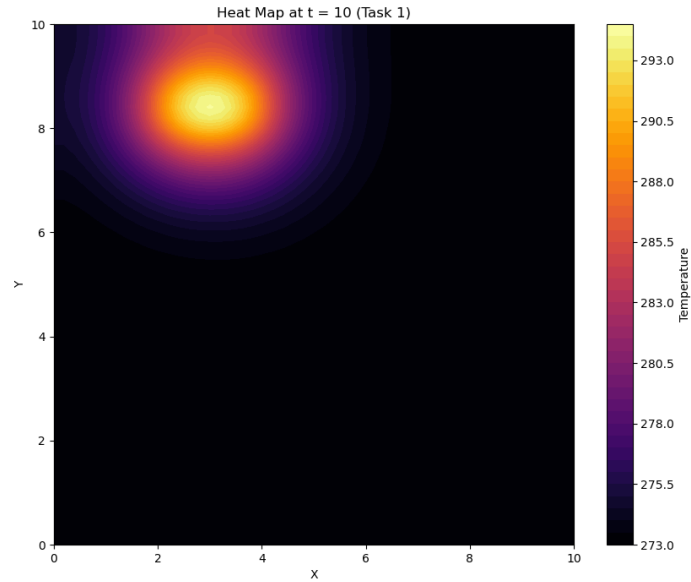


Figure 1: Heat map after 10 seconds. The heat from the source begins to propagate into the room.

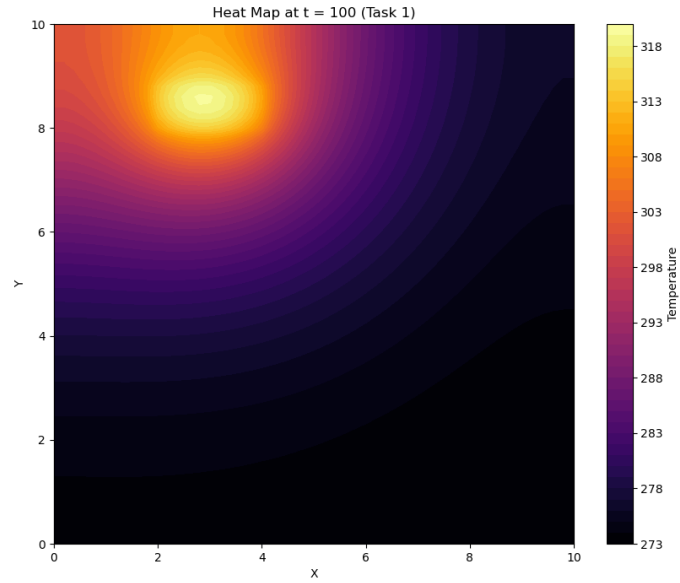


Figure 2: Heat map after 100 seconds. The diffusion process is well underway, with temperatures rising in the center.

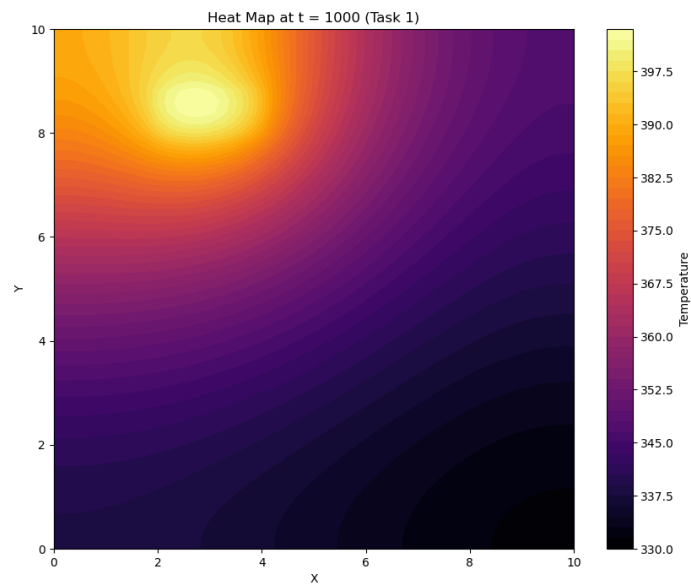


Figure 3: Heat map after 1000 seconds. The room's temperature is significantly elevated as the heater continues to add energy.

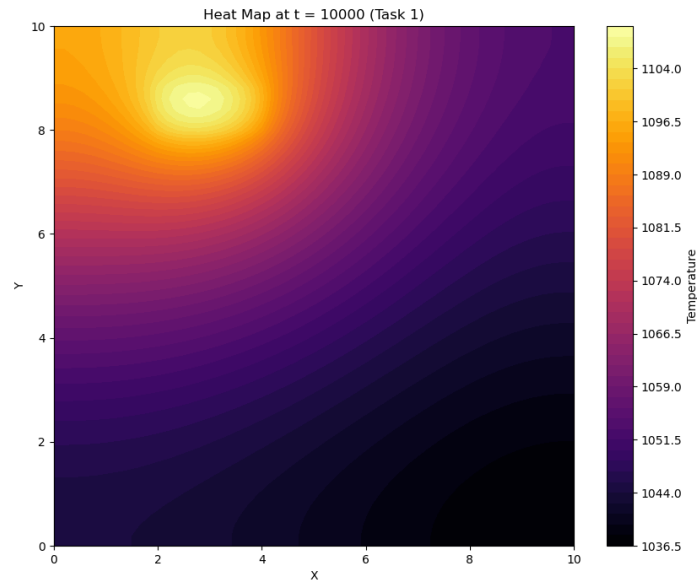


Figure 4: Heat map after 10000 seconds. The entire room has reached a very high temperature due to continuous heating without any heat loss.

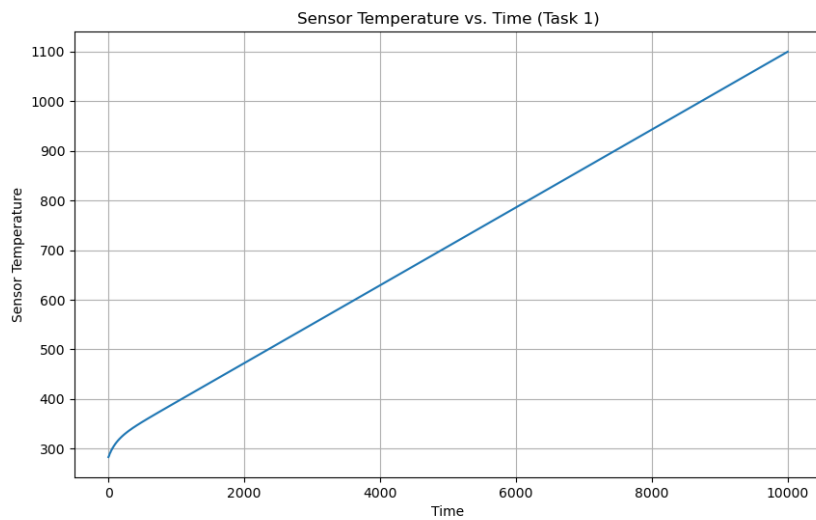


Figure 5: Sensor temperature evolution. The temperature shows a steady and unbounded increase, as expected in a perfectly insulated system with a constant heat source.

3.2 Scenario with Uniform Heat Loss

This scenario introduces heat loss through the walls and window with a uniform heat transfer coefficient ($h = h_w = 0.002$), representing a more realistic room.

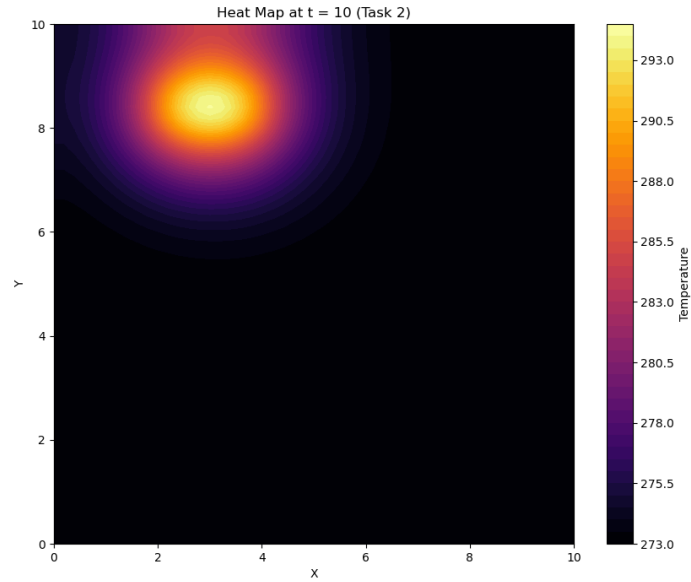


Figure 6: Heat map after 10 seconds. The initial heat spread is similar to the insulated case.

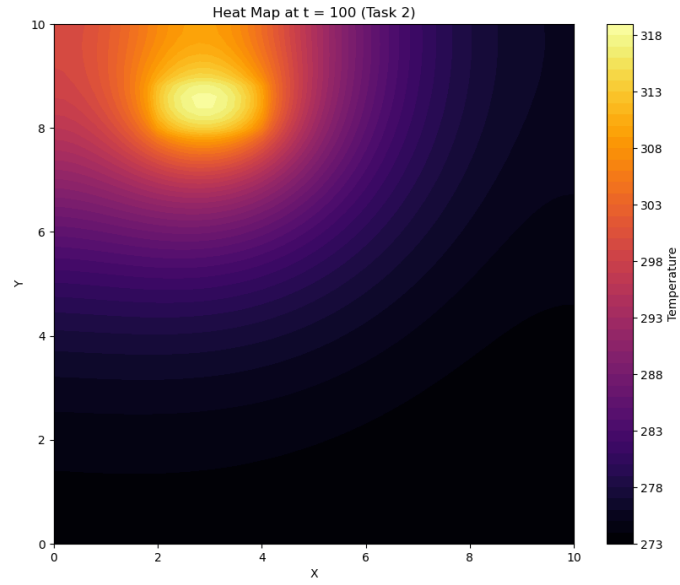


Figure 7: Heat map after 100 seconds. Heat loss through the boundaries begins to slow the overall temperature rise.

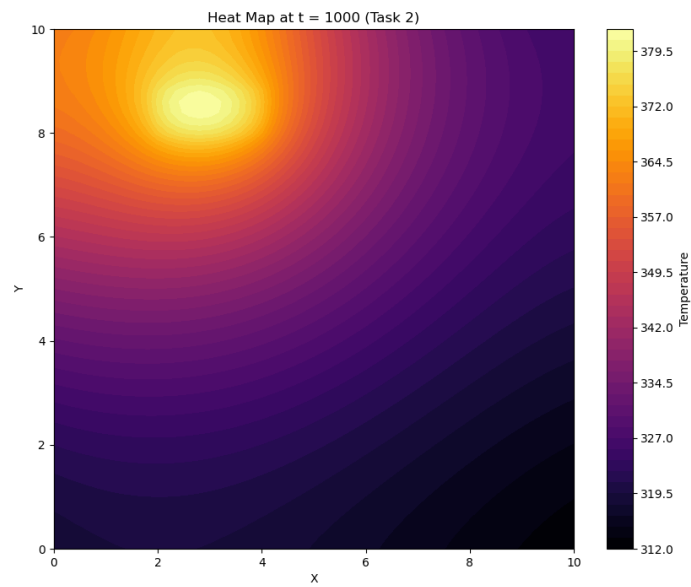


Figure 8: Heat map after 1000 seconds. The system is clearly trending towards a thermal equilibrium.

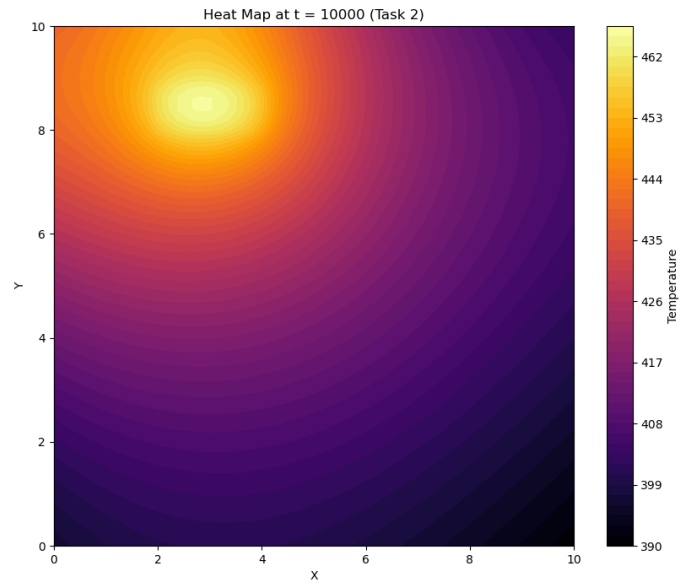


Figure 9: Heat map after 10000 seconds. A steady state is approached, where heat generated is balanced by heat loss.

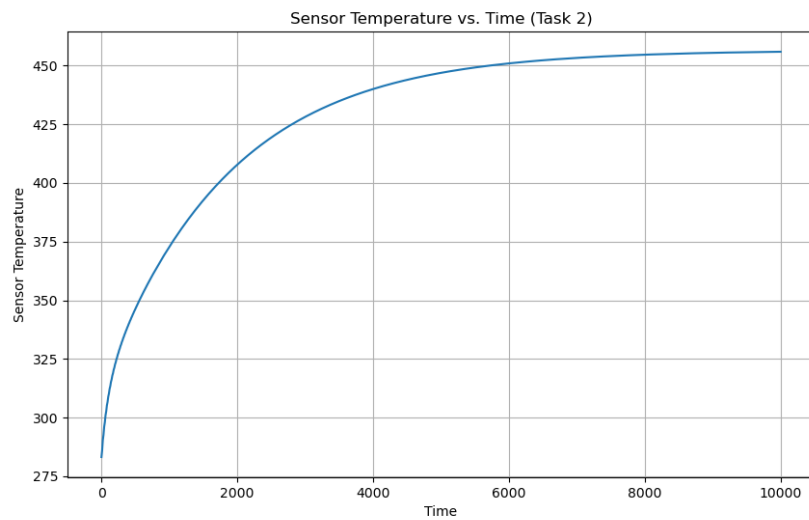


Figure 10: Sensor temperature evolution. The temperature rises and starts to plateau, indicating an approach towards thermal equilibrium with the environment.

3.3 Scenario with Dominant Window Heat Loss

Here, the heat loss through the window is significantly higher than through the walls ($h = 0.002, h_w = 0.01$), modeling a poorly insulated window.

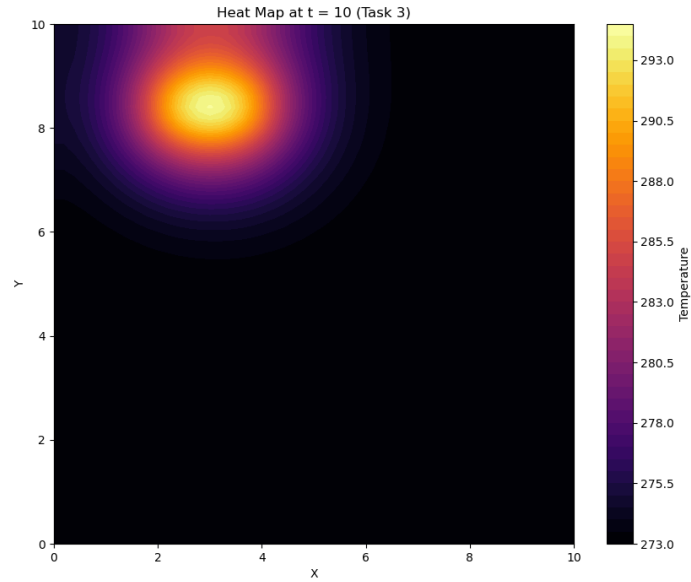


Figure 11: Heat map after 10 seconds. The initial phase of diffusion is unaffected by the boundary conditions.

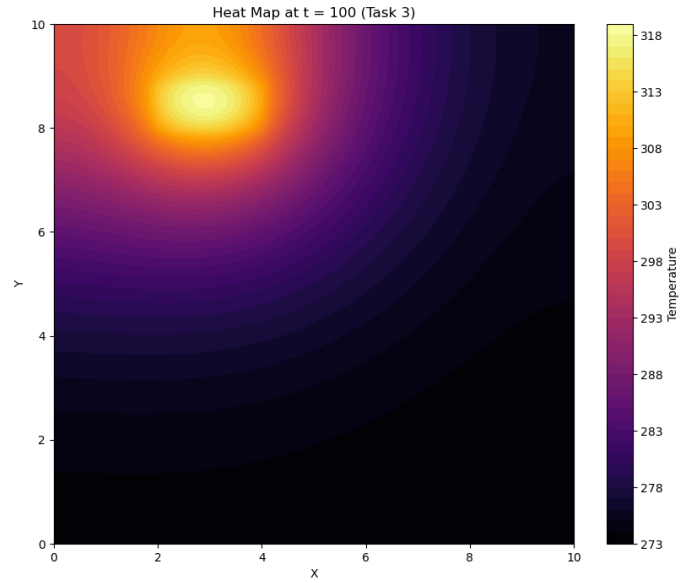


Figure 12: Heat map after 100 seconds. The cooling effect of the window on the right wall becomes apparent.

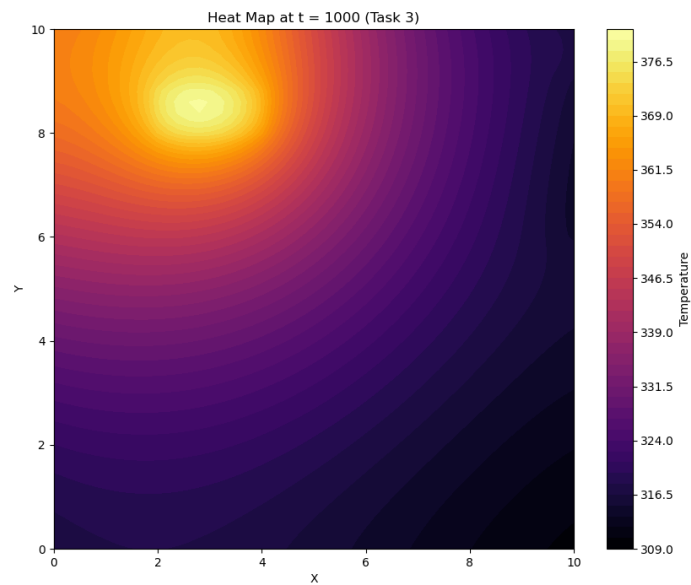


Figure 13: Heat map after 1000 seconds. A strong temperature gradient forms near the window, which acts as a major heat sink.

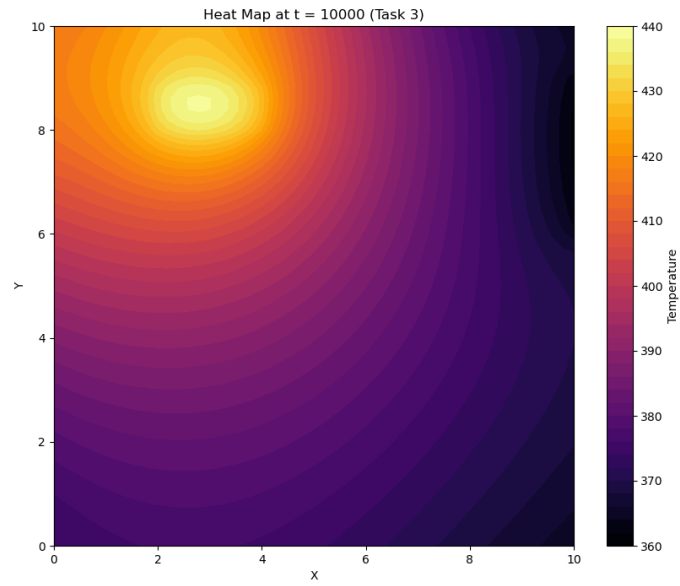


Figure 14: Heat map after 10000 seconds. The final steady state shows a distinctly cooler region near the window.

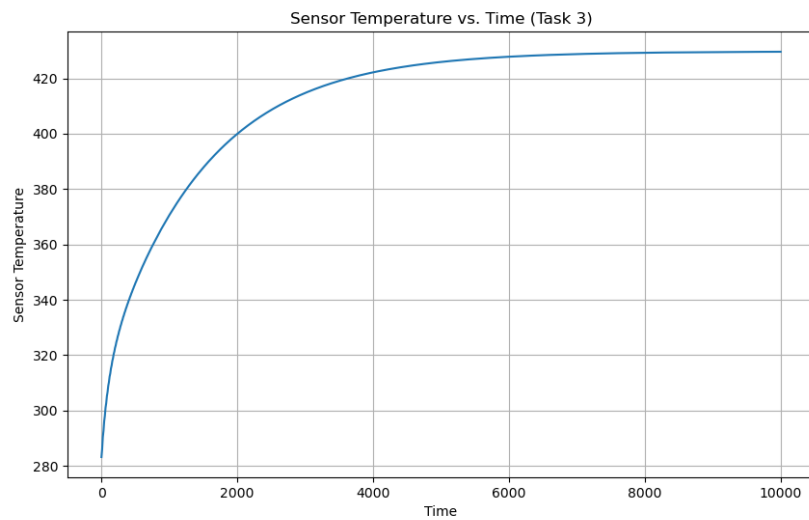


Figure 15: Sensor temperature evolution. The equilibrium temperature is lower than in the previous case, reflecting the increased rate of heat loss.

3.4 Scenario with Thermostat Control

This scenario is similar to the previous one but adds a thermostat that deactivates the heater when the sensor temperature exceeds 298K.

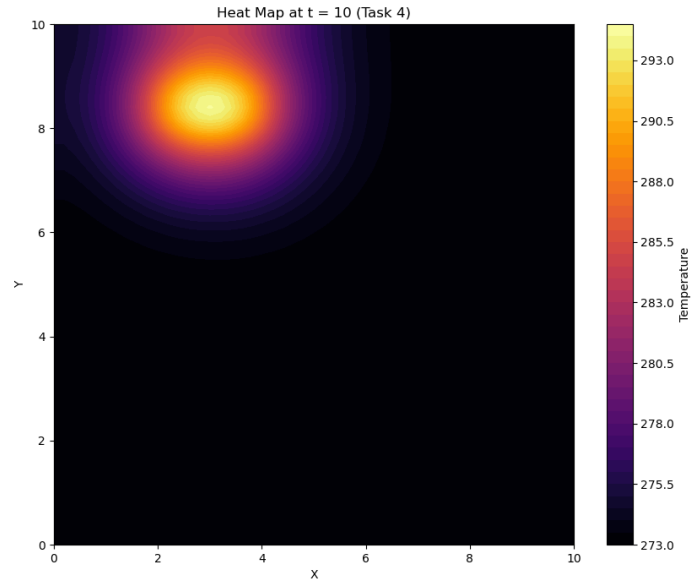


Figure 16: Heat map after 10 seconds. The heater is active during the initial warm-up phase.

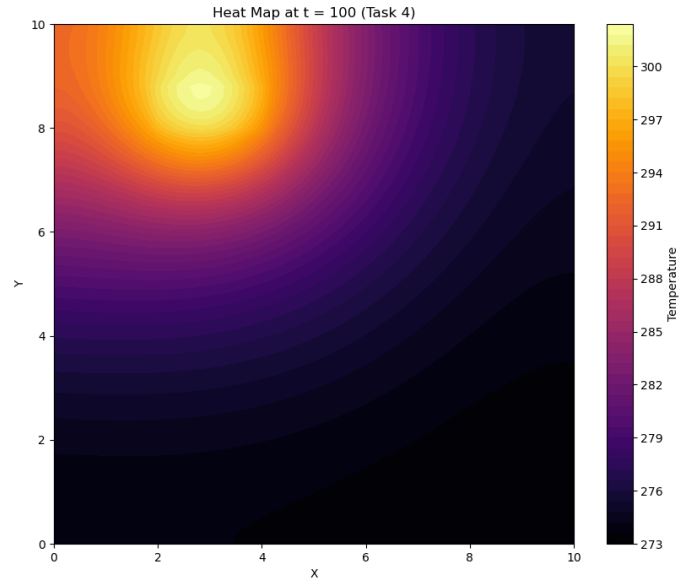


Figure 17: Heat map after 100 seconds. The temperature distribution reflects the heater's regulated, intermittent operation.

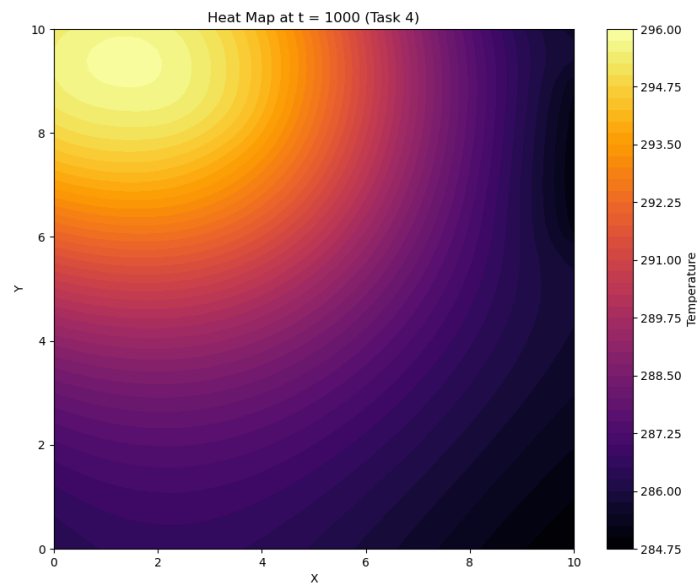


Figure 18: Heat map after 1000 seconds. The room temperature is maintained in a dynamic equilibrium around the setpoint.

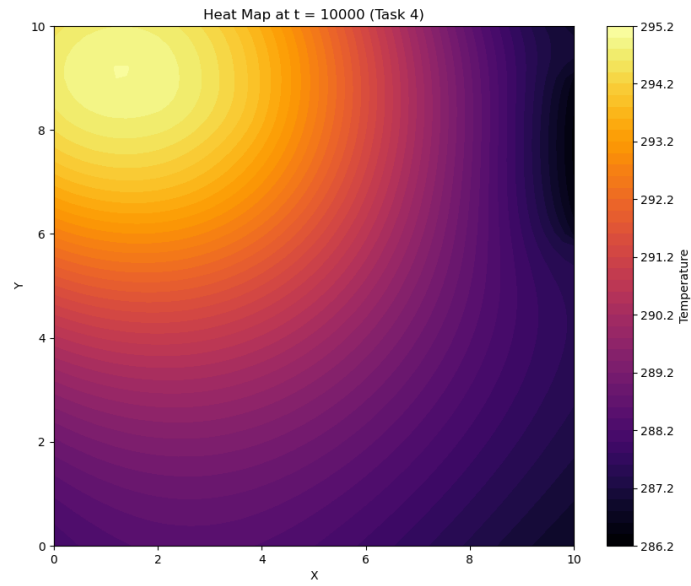


Figure 19: Heat map after 10000 seconds. The long-term average temperature is stable due to thermostat control.

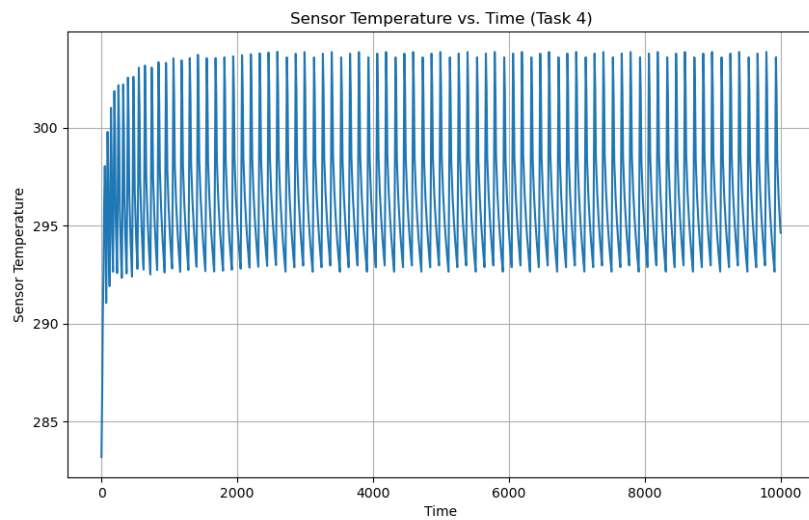


Figure 20: Sensor temperature evolution. The temperature oscillates around the thermostat's 298K setpoint, demonstrating the feedback control mechanism.

3.5 Energy Analysis with a Broken Window

This task models a "broken window" with a very high heat transfer coefficient ($h_w = 1.0$) and perfect insulation at the walls, with a thermostat active.

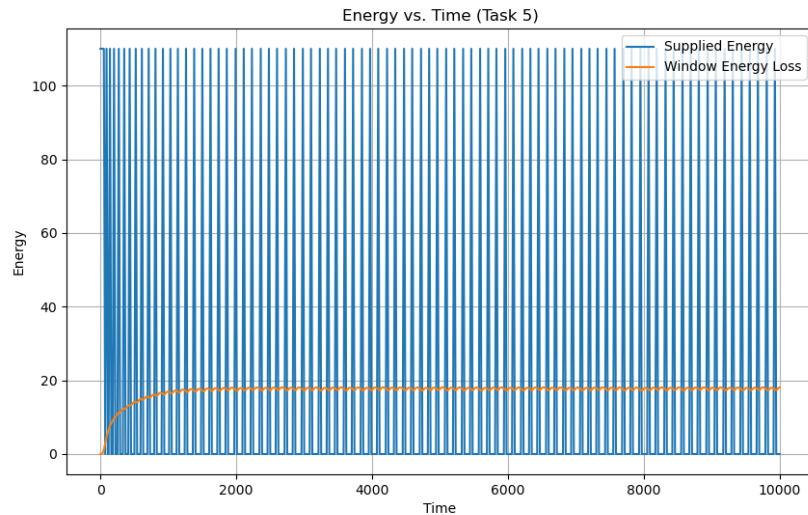


Figure 21: Supplied vs. Lost Energy. This plot compares the energy supplied by the heater to the energy lost through the broken window. The two should balance in a steady-state condition.

3.6 Energy Analysis with High External Temperature

This is a variation of the previous scenario where the external temperature is extremely high (10^4 K), reversing the typical heat flow direction.

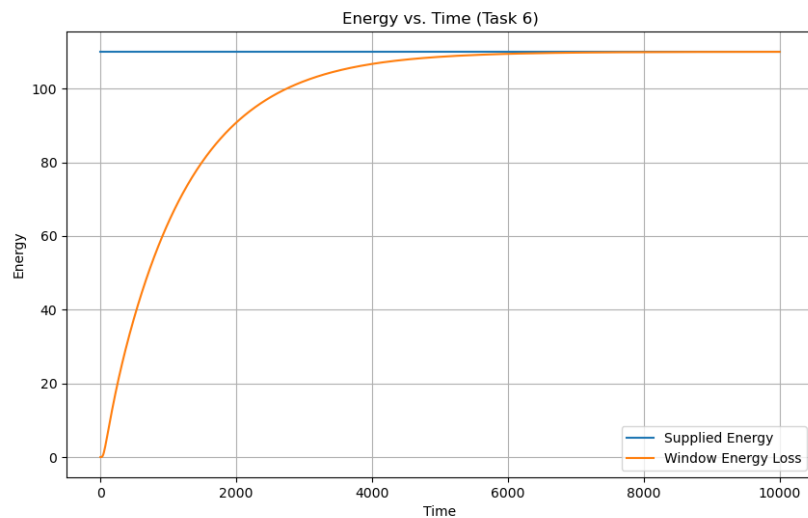


Figure 22: Supplied vs. Lost Energy. With a high external temperature, the "lost energy" becomes negative, indicating that a large amount of heat is flowing into the room from the outside.

4 Conclusions

This simulation project successfully modeled the process of 2D heat diffusion using the robust Crank-Nicolson numerical method. The results obtained from the six distinct tasks provide valuable insights into the behavior of the system under different physical constraints.

The first task, representing a perfectly isolated system, demonstrated the fundamental principle of energy conservation, where the temperature increased without bound due to the constant energy input. Tasks 2 and 3 illustrated the concept of thermal equilibrium, showing how the system's temperature stabilizes when heat loss to the environment is introduced. The temperature gradients observed, especially the pronounced effect of the window in Task 3, clearly visualize the path of heat flow from warmer to cooler regions.

Task 4 provided an excellent demonstration of a real-world feedback system. The oscillations of the sensor temperature around the thermostat's setpoint showcase the dynamic interplay between the heat source and the control system. This behavior is typical of thermostat-controlled heating systems.

Finally, Tasks 5 and 6 highlighted the energy dynamics of the system. The comparison of supplied and lost energy confirms the conservation of energy principle, where in a steady state, the energy injected by the heater is balanced by the energy escaping through the boundaries. The final task presented a non-intuitive scenario where the heat flow was reversed, demonstrating the algorithm's ability to handle generalized heat transfer problems.

Overall, the Crank-Nicolson method proved to be a stable and effective choice for this simulation. The results are physically plausible and provide a clear, visual understanding of the principles of heat transfer and feedback control.